J.A.R.V.I.S.

Just-in-time Agent for Research Validation & Ideation Support

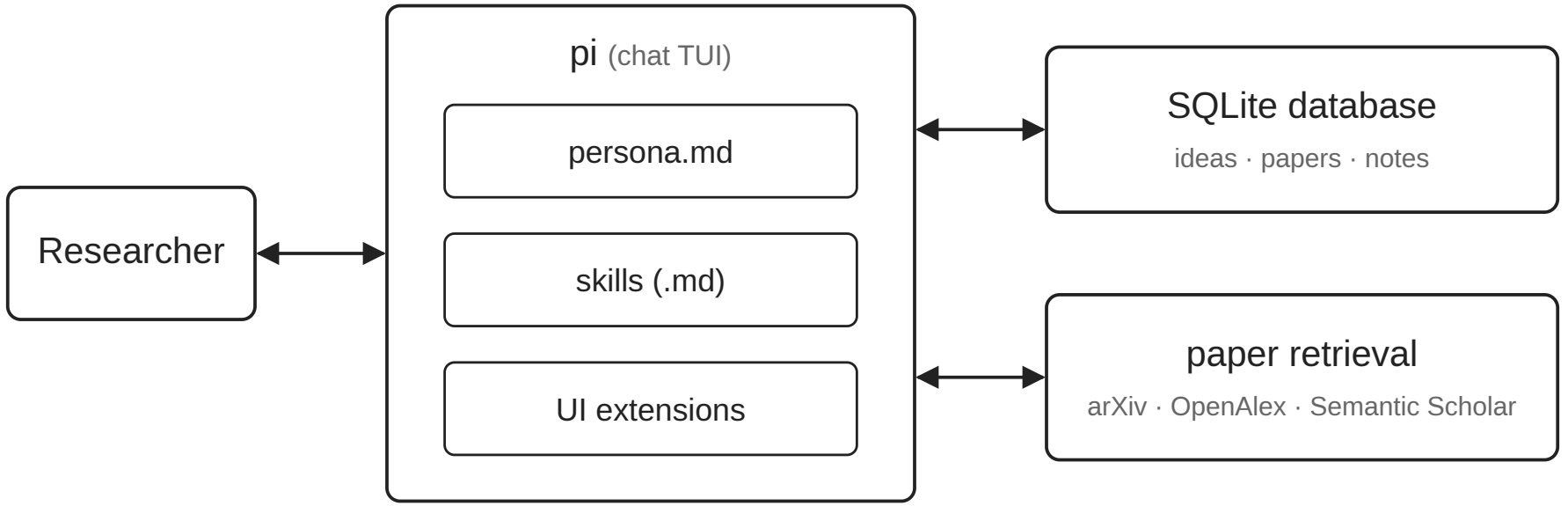
2026-10-27

The problem

- A researcher can have blind spots.
- It's easy to get stuck on a hard problem, and advisors can not be available at all times.

What J.A.R.V.I.S. does

It's a chat agent, like claude code.
Interface is built atop pi (minimal TUI coding assistant). Our work is composed of persona.md, .md skills, a sqlite database, ui extensions, and backend work for paper source retrieval.
Paper sources: arxiv, openalex, and semantic scholar.



Demo

I've read the project and saved a summary note. Where are you stuck, and what kind of next step do you want?

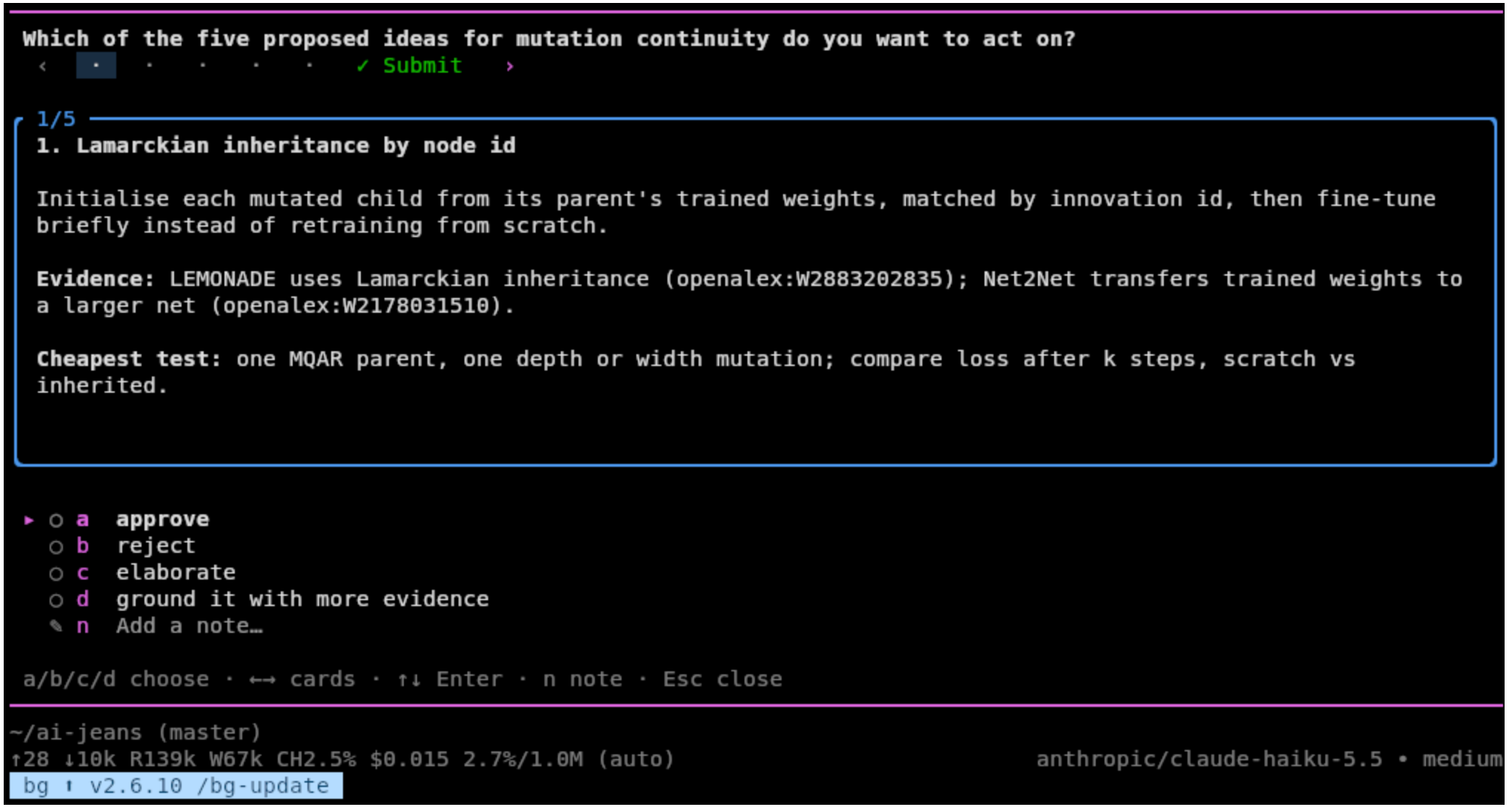
From the roadmap, the open problems I see are:

1. **No CNN→transformer path.** No mutation fills or empties a compute node's parameter slot, so evolution never gained a data×data matmul in 20 lineages × 60 generations.
2. **Mutation continuity.** Each child is retrained from scratch, with no parent weight transfer or output-distance bound.
3. **Coexistence protocol not yet run.** The frozen five-seed CNN/transformer run is still pending, and recurrence/diffusion work is gated on it.
4. **MAP-Elites is diagnostic only.** It never contributes parents to reproduction.

If one of these is what's blocking you, say which. If not, tell me the problem and the kind of step you want, for example: a cheap diagnostic, a literature sweep, or a new mechanism to try.

it's 2, I want to see how the industry solves this typically

Demo


Which of the five proposed ideas for mutation continuity do you want to act on?
‹ · · · · · ✓ Submit ›
1/5
1. Lamarckian inheritance by node id
Initialise each mutated child from its parent's trained weights, matched by innovation id, then fine-tune briefly instead of retraining from scratch.
Evidence: LEMONADE uses Lamarckian inheritance (openalex:W2883202835); Net2Net transfers trained weights to a larger net (openalex:W2178031510).
Cheapest test: one MQAR parent, one depth or width mutation; compare loss after k steps, scratch vs inherited.
▸ ○ a approve
○ b reject
○ c elaborate
○ d ground it with more evidence
✎ n Add a note…
a/b/c/d choose · ←→ cards · ↑↓ Enter · n note · Esc close
~/ai-jeans (master)
↑28 ↓10k R139k W67k CH2.5% $0.015 2.7%/1.0M (auto) anthropic/claude-haiku-5.5 • medium
bg ↑ v2.6.10 /bg-update

Preliminary results

- Because it's in idea-realm, I'm not concerned with accuracy moreso than breadth.
- The results, subjectively, have breadth as well as relevance.

Why it is creative

- Novelty: Dedup means that ideas progressively move on from what's already been seen.
- Ranking: Ideas are ranked by subective next-step promise in an Elo system.

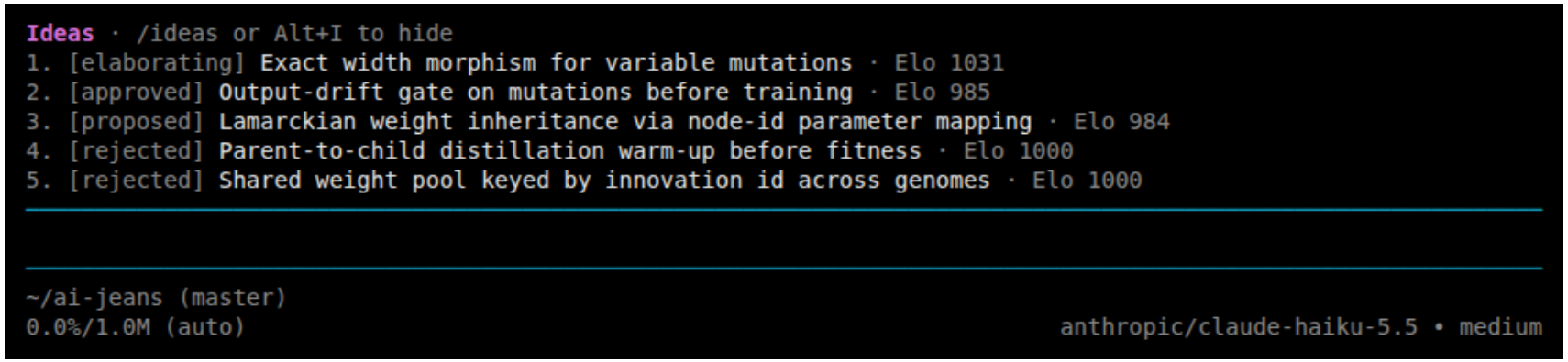
Deduplication

- Candidate new ideas are generated.
- Simple embeddings compare similarity to others in the database, rejecting too-similar new ideas.

We'll consider more sophisticated final judgements of uniqueness in another pass.

Ranking

- Ideas are ranked by Elo score.
- When an idea is accepted, 3 judge calls compare it to 3 other (semantically) nearby ideas.



Challenges

- addressed in previous slides

Work remaining

- Deeper eval of deduplication and ranking.